

Def.

Let  $V$  be a real or complex vector space.

A function  $f: V \times V \rightarrow F$  is called a sesqui-linear form if it satisfies that for all  $\alpha, \beta, \gamma \in V$  and  $c \in F$ ,

$$\begin{cases} f(c\alpha + \beta, \gamma) = c \cdot f(\alpha, \gamma) + f(\beta, \gamma) \\ f(\alpha, c\beta + \gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) \end{cases}$$

In this page, we call the  $f$  as form, for

Thm.

Let  $V$  be a finite dimensional inner product space and  $f: V \times V \rightarrow F$  be a form. Then, there is a unique linear operator  $T$  on  $V$  such that

$$f(\alpha, \beta) = \langle T\alpha, \beta \rangle \text{ for all } \alpha, \beta \in V.$$

Moreover, the map  $f \mapsto T$  is isomorphism from space of forms onto  $L(V, V)$ .

Proof. Let  $\beta \in V$  be fixed. Then, a map  $\alpha \mapsto f(\alpha, \beta)$  is a linear map on  $V$ .

By the Riesz-Representation theorem, for an orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ , there exists a unique vector  $\beta' = \overline{f(\alpha_1, \beta)} \cdot \alpha_1 + \dots + \overline{f(\alpha_n, \beta)} \cdot \alpha_n$  such that

$$\langle \alpha, \beta' \rangle = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Define  $U: V \rightarrow V; \beta \mapsto \beta'$ . Then, given  $\alpha, \beta, \gamma \in V$  and  $c \in F$ ,

$$\langle \alpha, U(c\beta + \gamma) \rangle = f(\alpha, c\beta + \gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) = \overline{c} \cdot \langle \alpha, U(\beta) \rangle + \langle \alpha, U(\gamma) \rangle = \langle \alpha, cU(\beta) + U(\gamma) \rangle$$

Since the  $\alpha$  was arbitrary, it proves  $U$  is linear.

Moreover, since  $V$  is finite-dimensional, so the adjoint exists, put  $T = U^*$ .

Consequently,  $f(\alpha, \beta) = \langle \alpha, U(\beta) \rangle = \langle T\alpha, \beta \rangle$ .  $\square$  Existence.

Corollary. Given forms  $f, g: V \times V \rightarrow F$ , define

$$\langle f, g \rangle \stackrel{\text{def}}{=} \text{Tr}(T_f T_g^*) \stackrel{(*)}{=} \sum_{j,k} f(\alpha_k, \alpha_j) \cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on  $V \times V$  such that  $(*)$ , for any orthonormal basis  $\{\alpha_1, \dots, \alpha_n\}$  for  $V$ .

Proof. Given orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ , let  $A = [T_f]_{\mathcal{B}}$  and  $B = [T_g]_{\mathcal{B}}$ .

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j \rangle = f(\alpha_k, \alpha_j) \text{ and } B_{jk} = \langle T_g(\alpha_k), \alpha_j \rangle = g(\alpha_k, \alpha_j).$$

Moreover,  $[T_f \circ T_g^*]_{\mathcal{B}} = [T_f]_{\mathcal{B}} \cdot [T_g^*]_{\mathcal{B}} = AB^*$ , so  $\text{Tr}(AB^*) = \sum_{j,k} A_{jk} \cdot \overline{B_{jk}}$  gives  $(*)$ .

Def. Let  $V$  be a fin. dim inner product space, with basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ .

For a given form  $f: V \times V \rightarrow F$ , the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of  $f$  in the ordered basis  $\mathcal{B}$ .

Observe that: for a form  $f: V \times V \rightarrow F$  and orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ .

Then,  $f\left(\sum_s \chi_s \alpha_s, \sum_r \gamma_r \alpha_r\right) = \sum_{r,s} \overline{\gamma_r} A_{rs} \chi_s$ , or

$$f(\alpha, \beta) = Y^* A X \text{ where } X = [\alpha]_{\mathcal{B}}, Y = [\beta]_{\mathcal{B}}.$$

Thm. Let  $V$  be a fin. dim Complex inn. Prod. space.

For any form on  $V \times V$ , there is an orthonormal basis for  $V$ ,

in which the matrix of  $f$  is upper-triangular.

Proof. Since  $V$  is fin. dim. Complex. inn. Prod. space, for linear operator  $T: V \rightarrow V$  s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle,$$

it has an orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  for  $V$  s.t.  $[T]_{\mathcal{B}}$  is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = 0 \text{ if } j > k.$$

Def. A form  $f$  on  $V$  is called Hermitian if

$$f(\alpha, \beta) = \overline{f(\beta, \alpha)} \quad \text{for all } \alpha, \beta \in V.$$

Note: if there is a linear operator  $T$  on  $V$  such that  $f(\alpha, \beta) = \langle T(\alpha), \beta \rangle$ , then  $\overline{f(\beta, \alpha)} = \langle \alpha, T(\beta) \rangle = \langle T^*(\alpha), \beta \rangle$ , so,

$f$  is Hermitian if and only if  $T$  is Hermitian.

If  $T$  is Hermitian, then  $\langle T(\alpha), \alpha \rangle$  is real for all  $\alpha \in V$ , so  $f(\alpha, \alpha)$  is real.

Lemma. Let  $V$  be a Complex inner product space, and  $f$  is a form on  $V$  such that  $f(\alpha, \alpha)$  is real, for all  $\alpha \in V$ .

Then,  $f$  is Hermitian.

Proof. Given  $\alpha, \beta \in V$ ,

$$f(\alpha + \beta, \alpha + \beta) = f(\alpha, \alpha) + f(\beta, \beta) + f(\alpha, \beta) + f(\beta, \alpha)$$

$$f(\alpha + i\beta, \alpha + i\beta) = f(\alpha, \alpha) + f(i\beta, i\beta) - i \cdot f(\alpha, \beta) + i f(\beta, \alpha)$$

By the assumption,  $f(\alpha, \beta) + f(\beta, \alpha)$  and  $-i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha)$  are real. Thus,

$$f(\alpha, \beta) + f(\beta, \alpha) = \overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)} \quad \text{and} \quad -i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha) = i \cdot \overline{f(\alpha, \beta)} - i \cdot \overline{f(\beta, \alpha)}$$

Multiplying  $i$  to the right, then we obtain

$$f(\alpha, \beta) - f(\beta, \alpha) = -\overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)}$$

so,  $2f(\alpha, \beta) = 2 \cdot \overline{f(\alpha, \beta)}$ , or  $f$  is Hermitian.  $\square$

Combining the above note with this Lemma if  $V$  is fin. dim, then

$T$  is Hermitian if and only if  $\langle T(\alpha), \alpha \rangle$  is real.

### Principal Axis Theorem.

Thm. For every Hermitian form  $f$  on a fin. dim inn. prod space  $V$ , there exists an orthonormal basis  $\beta = \{\alpha_1, \dots, \alpha_n\}$  such that

$f$  is represented by a diagonal matrix with real entries.

Proof. Since  $V$  is fin. dim, there exists a linear operator  $T$  on  $V$  s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above,  $f$  is Hermitian iff  $T$  is Hermitian, so there is an orthonormal basis for  $V$  which is consisted by eigenvector of  $T$ . Denote  $\beta = \{\alpha_1, \dots, \alpha_n\}$ , and  $C_i$  be the eigenvalue corresponding to  $\alpha_i$ . Then,

$$f(\alpha_k, \alpha_i) = \langle T(\alpha_k), \alpha_i \rangle = C_k \cdot \delta_{ki},$$

this proved. ( $C_k$  are real by  $T$  is Hermitian). ~~Q~~

Note: Under the condition hold,

$$f\left(\sum x_i \alpha_i, \sum y_k \alpha_k\right) = \sum_i C_i x_i \overline{y_i}$$

THK. 9.2. (324P)

Def.

A sesqui-linear form  $f$  is said to be non-degenerate if: for fixed  $\alpha \in V$ ,  
if  $f(\alpha, \beta) = 0$  for all  $\beta \in V$ , then  $\alpha = 0$ .

Prop.

Let  $f: V \times V \rightarrow F$  be a form on an inner product space  $V$ .

$f$  is non-degenerate if and only if the associated linear  $T_f$  is non-singular.

Proof. Suppose that  $f$  is non-degenerate. To show  $T_f$  is non-singular, ( $\ker T_f = \{0\}$ )  
assume that  $T_f(\alpha) = 0$ . That is, for any  $\beta \in V$ ,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle = \langle 0, \beta \rangle = 0$$

So,  $\alpha = 0$  by definition. Conversely, if  $T_f$  is non-singular, then for any non-zero  $\alpha \in V$ ,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle \neq 0 \text{ if } \beta = T_f(\alpha)$$

which proves  $f$  is non-degenerate,  $\square$

Prop.

Let  $f: V \times V \rightarrow F$  be a non-degenerate form on a finite dimensional <sup>inn. prod</sup> space  $V$ .  
Given linear functional  $L$ , there exists one and only one vector  $\beta \in V$  such that  
$$L(\alpha) = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Proof By the theorem above,  $f$  has invertible associated linear map  $T$  such that  
$$f(\alpha, \gamma) = \langle T(\alpha), \gamma \rangle \text{ for all } \alpha, \gamma \in V.$$

Meanwhile, by the Riesz Representation theorem, for the given  $L$ ,  
there exists a unique vector  $\beta' = L(\alpha_1) \cdot \alpha_1 + \dots + L(\alpha_n) \alpha_n$ , ( $\{\alpha_1, \dots, \alpha_n\}$  is orthonormal)  
$$L(\alpha) = \langle \alpha, \beta' \rangle \text{ for all } \alpha \in V.$$

Now, let  $\beta = (T^*)^{-1}(\beta')$ , then

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle = \langle T(\alpha), (T^*)^{-1}(\beta') \rangle = \langle \alpha, (T^*) \circ (T^*)^{-1}(\beta') \rangle = \langle \alpha, \beta' \rangle = L(\alpha).$$

And, the uniqueness is given by:  $f(\alpha, \beta_1) = L(\alpha) = f(\alpha, \beta_2) \Rightarrow f(\alpha, \beta_1 - \beta_2) = 0$  and  
Since  $f$  is non-degenerate, so  $\beta_1 - \beta_2$  must be zero. (Let  $\alpha = \beta_1 - \beta_2$ ) ~~no~~

Prop.

Let  $f: V \times V \rightarrow F$  be a non-degenerate form on a finite dimensional inner Product Space  $V$ .  
Given a linear operator  $S: V \rightarrow V$ , there exists an adjoint  $S'$  relative to  $f$ .  
(That is,  $f(S\alpha, \beta) = f(\alpha, S'\beta)$  for all  $\alpha, \beta \in V$ ).

Proof